

GREYTOWN, South Africa

WMO: 684870

Lat: 29.083S

Lon: 30.603E

Elev: 3376

StdP: 12.99

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.6	37.0	21.6	18.3	62.0	25.4	22.0	59.9	16.2	60.4	14.2	60.5	1.4	10	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.8	90.8	67.4	87.9	67.0	85.1	66.4	73.5	83.2	71.9	80.9	70.4	79.0	7.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	129.6	78.1	69.3	122.4	76.1	68.0	116.7	74.5	39.7	82.9	38.1	80.8	36.7	78.9	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.3	11.1	DB	29.8	96.7	2.5	3.2	28.0	99.0	26.5	100.9	25.2	102.6	23.4	104.9	(4)
(5)			WB	29.0	76.4	2.5	2.6	27.2	78.3	25.8	79.8	24.4	81.3	22.6	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	70.1	70.8	69.2	64.5	59.4	55.2	54.8	59.2	62.5	64.2	66.4	69.0	(6)	
	DBStd	8.20	5.78	4.91	5.28	5.55	5.57	5.37	5.52	6.76	7.81	7.52	6.92	6.34	(7)	
	HDD50	48	0	0	0	0	3	12	18	8	5	1	1	0	(8)	
	HDD65	1475	21	10	23	74	186	297	316	202	138	109	66	33	(9)	
	CDD50	5064	623	582	596	435	296	168	168	293	378	443	493	589	(10)	
	CDD65	1016	179	172	154	59	13	3	1	22	61	86	109	157	(11)	
	CDH74	7135	1207	952	895	344	156	33	29	222	610	750	798	1139	(12)	
	CDH80	2213	414	284	257	67	24	2	2	36	193	264	261	409	(13)	
(14) Wind	WSAvg	4.2	4.5	4.1	3.7	3.4	3.2	3.4	3.6	4.5	4.8	5.1	5.3	4.8	(14)	
(15) Precipitation	PrecAvg	31.1	4.6	4.1	3.6	1.5	0.7	0.6	0.9	1.1	1.7	3.4	4.1	5.0	(15)	
	PrecMax	41.8	9.0	10.0	6.1	4.4	2.6	3.7	5.5	3.8	4.6	8.1	8.3	8.3	(16)	
	PrecMin	19.1	1.4	1.5	1.9	0.3	0.0	0.0	0.0	0.0	0.2	0.7	1.3	2.1	(17)	
	PrecStd	5.7	2.0	2.2	1.3	1.0	0.7	0.8	1.5	0.9	1.1	1.8	1.6	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	91.7	89.9	85.8	83.2	77.8	77.8	84.2	90.8	92.7	91.7	94.4	(19)	
		MCWB	70.5	70.1	69.2	65.0	61.2	55.5	53.7	56.5	61.3	63.8	65.2	71.7	(20)	
	2%	DB	88.6	87.0	86.1	81.4	78.3	73.3	73.4	79.6	85.4	87.1	87.0	88.5	(21)	
		MCWB	69.7	70.0	68.2	63.9	59.4	53.9	52.2	55.7	59.4	62.5	66.0	69.0	(22)	
	5%	DB	84.5	83.4	82.7	77.8	74.2	69.6	70.0	75.9	80.8	82.4	82.8	84.4	(23)	
		MCWB	69.3	69.5	67.7	63.1	58.0	52.5	51.9	54.7	58.9	62.3	65.5	68.6	(24)	
	10%	DB	80.7	79.9	79.1	74.4	70.5	66.1	66.2	71.8	76.5	77.8	78.5	80.4	(25)	
		MCWB	68.8	69.1	67.3	62.5	56.7	51.5	51.2	54.0	58.1	61.9	65.1	67.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	75.4	73.8	70.5	64.8	60.7	57.6	62.4	66.4	69.0	71.9	74.9	(27)	
		MCDB	85.7	85.2	82.9	78.6	75.6	68.3	67.3	73.0	81.4	83.5	82.3	87.8	(28)	
	2%	WB	72.9	73.3	71.4	67.3	62.3	57.0	55.7	59.8	63.3	66.5	69.5	72.4	(29)	
		MCDB	82.3	81.5	80.6	75.0	72.2	67.7	65.6	71.0	77.2	78.7	79.9	83.2	(30)	
	5%	WB	70.9	71.6	69.7	65.4	60.3	55.0	54.3	58.0	61.4	64.8	67.9	70.4	(31)	
		MCDB	79.7	78.9	78.0	73.0	70.1	64.6	64.0	68.3	73.7	76.3	77.7	80.1	(32)	
	10%	WB	69.3	70.0	68.3	63.9	58.5	53.4	52.8	56.4	59.8	63.3	66.4	68.7	(33)	
		MCDB	77.9	77.0	76.1	71.0	67.1	62.5	62.3	66.1	70.9	73.7	75.3	77.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	16.8	18.0	19.4	23.9	24.1	24.6	22.7	22.2	19.2	18.1	17.8	(35)	
		MCDBR	25.5	23.4	24.4	25.4	29.1	28.0	28.6	28.7	30.3	29.8	26.6	26.2	(36)	
	5% WB	MCWBR	9.9	8.8	9.5	9.6	12.2	12.1	12.5	10.1	9.9	9.8	9.8	10.2	(37)	
		MCWBR	21.8	20.1	21.4	20.3	25.5	22.9	24.2	22.5	25.4	24.7	22.6	22.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.394	0.385	0.377	0.347	0.319	0.302	0.310	0.367	0.412	0.432	0.403	0.400	(40)	
		taud	2.368	2.395	2.400	2.443	2.485	2.508	2.474	2.317	2.216	2.203	2.314	2.340	(41)	
	Ebn at Noon		301	298	290	283	277	275	277	271	274	280	296	300	(42)	
		Edh at Noon	42	39	37	32	28	25	28	36	44	47	44	43	(43)	
(44) All-Sky Solar Radiation	RadAvg		1894	1863	1643	1359	1188	1067	1143	1331	1528	1614	1786	1898	(44)	
	RadStd		138	149	91	85	64	55	65	83	95	142	153	118	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page